

Observable-Class Mismatch and the Corrected $\delta \mapsto \beta^*$ Relation on Heisenberg Graphs: Block Normalisation, Central-Phase Contribution, and Structural Correspondence

Jérôme Beau

2026

Abstract

Papers O12–O13 [17, 18] established that the exact Weil-block capacity exponent $\hat{\delta}_{\text{exact}}(q)$ decreases monotonically from 4.80 to 3.59 as q ranges over $\{29, 61, 101, 151, 211\}$, refuting the finite-size hypothesis S1 and ruling out convergence toward the phenomenological target $\delta \in [7.4, 10.6]$. The tension between $\hat{\delta}_{\text{exact}}$ and the window $\beta^* \in (0.09, 0.13)$ [8] is therefore not a numerical artefact but a structural mismatch identified as Source S2 in O13: the proxy-level derivation of the $\delta \mapsto \beta^*$ relation [12] does not account for the block-normalisation by $|S_n|$, the internal block dimension q , or the central-phase contribution $\psi_c(\gamma + bx)$ present in every exact Weil fingerprint. This paper resolves S2. We identify the observable-class mismatch precisely: the O7 derivation [12] operates on a scalar global capacity, while $\hat{\delta}_{\text{exact}}$ is extracted from a block-by-block mean over representations of internal dimension q . We decompose the exact observable as $\Sigma_n^{(c)} = \Sigma_n^{\text{geom}} + \Sigma_n^\gamma$, derive an effective exponent $\delta_{\text{eff}}(q)$ that absorbs the central-phase contribution, and obtain the corrected structural relation

$$\beta^* = \frac{1}{\delta_{\text{eff}} + \frac{1}{2}} + \epsilon(q),$$

where $\epsilon(q) \rightarrow 0$ as $q \rightarrow \infty$. Numerical re-fits of δ_{eff} at $q \in \{29, 61, 101, 151, 211\}$ determine which of two scenarios is realised: S2-A (reconciliation, β^* recovered in $(0.09, 0.13)$) or S2-B (structural revision of the amplification mechanism of O3 [8]).

Keywords. Heisenberg graphs, exact Weil-block capacity, projective capacity, corrected $\delta \rightarrow \beta^*$ relation, block normalisation, central-phase bias, spectral admissibility, phase coherence, shell-level versus block-level observables.

Contents

1	Introduction	3
1.1	Position in the O-series	3
1.2	Logical status of the three central objects	3
1.3	Main results	4
2	Background and Notation	4
2.1	Heisenberg graph and Weil blocks	4
2.2	Proxy-level capacity and the O7 derivation	4
3	Observable-Class Mismatch	5
3.1	Failure of assumption (A1): block heterogeneity	5
3.2	Failure of assumption (A2): normalisation by q	5
3.3	Failure of assumption (A3): central-phase contribution	6
4	Decomposition of the Exact Observable	6
5	Corrected Structural Relation	7
6	Pipeline-Based Numerical Check of the Corrected Relation	7
6.1	Status of this section	7
6.2	Computation of corrections	7
6.3	Derivation of the block-normalisation exponent $\eta = 1/2$	8
6.4	Connection between δ_γ , $\epsilon(q)$, and phase variance	9
6.5	Pipeline validation results and interpretation	10
6.6	Two scenarios	11
7	Physical Consequences	11
7.1	Scenario S2-A: programme consistency	11
7.2	Scenario S2-B: required revisions	12
7.3	Quark and neutrino sectors	12
8	Open Directions	12
9	Conclusion	13

1 Introduction

The Cosmochrony framework [1, 2] derives particle-physics observables from spectral properties of relational graphs associated with binary polyhedral subgroups of $SU(2)$. Within the admissibility sub-programme, a central chain of papers—O3 through O7 [8, 9, 10, 11, 12]—derives the relation

$$\beta^* = \frac{1}{\delta + \frac{1}{2}}, \quad (1)$$

linking the cascade exponent β^* to the capacity decay exponent δ . Charged-lepton mass ratios constrain $\beta^* \in (0.09, 0.13)$ [8], which via (1) requires $\delta \in [7.4, 10.6]$.

Papers O8–O13 [13, 14, 15, 16, 17, 18] develop the computational machinery to measure δ on Heisenberg graphs $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ via exact Weil-block projective capacity. O13 [18] establishes that $\hat{\delta}_{\text{exact}}(q)$ decreases monotonically with q , with no extrapolation of the five-prime dataset producing $\delta_\infty > 5.0$. The finite-size hypothesis S1 (“larger primes will push $\hat{\delta}_{\text{exact}}$ into $[7.4, 10.6]$ ”) is ruled out.

O13 identifies Source S2 as the remaining explanation of the tension: the structural relation (1) was derived at the proxy level and may not apply without modification to the exact Weil-block observable. Two specific gaps are named in O13, Section 8.3:

1. **Normalisation gap.** $\Sigma_n^{(c)}$ is normalised by the BFS shell size $|S_n|$, not by the block dimension q . The O7 derivation absorbs this normalisation at the proxy level; its effect in the exact-block setting is not derived.
2. **Central-phase contribution.** The fingerprint phase $\psi_c(\gamma + bx)$ mixes the central coordinate γ and the lateral coordinate b in a block-dependent way. The average over generic blocks c that defines $\hat{\delta}_{\text{exact}}$ carries a contribution from this phase that is absent from the proxy-level derivation of (1).

The present paper resolves S2 by re-deriving the $\delta \mapsto \beta^*$ relation in the exact-block setting. **S1 is not revisited here:** it was closed definitively by O13, and all data and arguments pertaining to S1 are taken as established. O14 has a single logical target: explaining why the proxy-level relation (1) changes class when δ is replaced by $\hat{\delta}_{\text{exact}}$, and deriving the corrected relation that holds at the exact-block level.

1.1 Position in the O-series

The O-series maps as follows: O11 (proxy observable [16]), O12 (exact observable [17]), O13 (asymptotic stability and closure of S1 [18]), O14 (present paper: theoretical correspondence, closure of S2). O14 does not add numerical measurements; it supplies the structural theory that was identified as missing in O13 §8.3.

1.2 Logical status of the three central objects

Three objects play distinct logical roles throughout this paper, and their hierarchy must be kept clear.

$\hat{\delta}_{\text{exact}}$ The *measured observable*: the power-law exponent extracted by log-log regression from the mean block capacity $\bar{\Sigma}_n$ on G_q , as defined and measured in O12–O13 [17, 18]. It is an empirical quantity associated with a specific computational procedure (exact Weil projection, BFS on G_q , fitting window satisfying (E1')–(E3)). Its values $\hat{\delta}_{\text{exact}}(q) \in [3.59, 4.80]$ for $q \in \{29, 61, 101, 151, 211\}$ are inputs to O14, not outputs.

δ_{eff} The *corrected observable*: $\hat{\delta}_{\text{exact}}$ after subtraction of the central-phase bias $\delta_{\gamma, \text{corr}}$ and the normalisation correction $\eta \cdot \log q / \log n^*$. It is the exponent that enters the structural relation (1) once the observable-class mismatch has been accounted for. The normalisation exponent η is a *benchmark parameter*: its central value $\eta = 1/2$ is used in all numerical tables, but its analytical derivation from the metaplectic phase structure of ρ_c is open direction O14-O1 (Section 8).

β^* The *inferred structural parameter*: the cascade exponent entering the mass-ratio formula of O3 [8], constrained phenomenologically to $\beta^* \in (0.09, 0.13)$. It is not directly measured; it is inferred from δ_{eff} via the corrected relation (7). Whether β^* falls in $(0.09, 0.13)$ after correction determines which scenario (S2-A or S2-B) is realised.

The distinction between $\hat{\delta}_{\text{exact}}$ (measured, procedure-dependent) and δ_{eff} (theory-corrected, procedure-independent in the $q \rightarrow \infty$ limit) is the central conceptual contribution of O14.

1.3 Main results

1. We exhibit the precise class of observable implicit in the O7 derivation and show it differs from the exact-block observable by two explicit corrections (Section 3).
2. We derive the decomposition $\hat{\delta}_{\text{exact}} = \delta_{\text{geom}} + \delta_{\gamma}$, where δ_{γ} is the central-phase contribution (Section 4).
3. We derive the corrected structural relation $\beta^* = 1/(\delta_{\text{eff}} + 1/2) + \epsilon(q)$ and show $\epsilon(q) \rightarrow 0$ as $q \rightarrow \infty$ (Section 5).
4. We re-fit δ_{eff} numerically and determine which scenario (S2-A or S2-B) is realised (Section 6).

2 Background and Notation

2.1 Heisenberg graph and Weil blocks

Fix an odd prime q . The Heisenberg group $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ has order q^3 and centre $Z(G_q) \cong \mathbb{Z}/q\mathbb{Z}$. An element is written (a, b, γ) with $a, b \in \mathbb{Z}/q\mathbb{Z}$ and $\gamma \in \mathbb{Z}/q\mathbb{Z}$. The $q-1$ faithful irreducible representations ρ_c ($c \in (\mathbb{Z}/q\mathbb{Z})^\times$) each have dimension q .

A *Weil block* associated with c is the set of Cayley graph generators $\{g : \rho_c(g) \neq 0\}$ acting on the representation space $V_c \cong \mathbb{C}^q$. The *block-level projective capacity* at BFS depth n is

$$\Sigma_n^{(c)} = \frac{\Delta r_n^{(c)}}{|S_n|}, \quad (2)$$

where $\Delta r_n^{(c)}$ is the incremental projective span in the ρ_c -representation and $|S_n|$ is the BFS shell size at depth n . The mean over blocks is $\bar{\Sigma}_n = \mathbb{E}_c[\Sigma_n^{(c)}]$.

2.2 Proxy-level capacity and the O7 derivation

The O7 derivation [12] operates on a *scalar global capacity* C_n^{proxy} , defined via a Fourier-character proxy that abelianises the representation structure. The derivation yields (1) under three implicit assumptions:

- (A1) The observable is a *scalar* function of BFS depth n , independent of the representation label c .
- (A2) The normalisation is *homogeneous*: the only length scale entering the normalisation is $|S_n|$.
- (A3) There is no *internal structure* within the observable: no block dimension q , no phase mixing between γ and b .

All three assumptions hold for the Fourier-character proxy of O11 [16]. None of them holds for the exact Weil observable $\Sigma_n^{(c)}$ measured in O12–O13 [17, 18]. The failure of (A1)–(A3) in the exact setting constitutes the observable-class mismatch whose consequences we now make precise.

3 Observable-Class Mismatch

3.1 Failure of assumption (A1): block heterogeneity

For the exact Weil observable, $\Sigma_n^{(c)}$ varies across blocks c at fixed n . O12 Table 2 and O13 Table 3 [17, 18] show that condition (E2), which requires block-by-block universality, is violated at $q \in \{29, 53, 61\}$ and satisfied only at $q \in \{101, 151, 211\}$ after the representation variety has been sufficiently sampled.

Consequently, $\hat{\delta}_{\text{exact}}$ is a property of the *mean* $\bar{\Sigma}_n$, not of an individual scalar observable. The O7 derivation operates on the latter; its extension to the former requires an averaging argument whose validity is established here for the first time.

Proposition 1 (Mean universality suffices). *Suppose $\bar{\Sigma}_n \sim C_0 n^{-\delta_{\text{eff}}}$ holds with $R^2 \geq 0.97$ over a fitting window satisfying condition (E1'). Then the structural relation (1) applies to δ_{eff} in place of δ , provided the variance correction $\epsilon(q)$ defined in Section 5 is included.*

Proof. The O7 derivation requires only that the capacity observable decay as a power law with a well-defined exponent. Mean universality supplies this requirement for $\bar{\Sigma}_n$. The correction $\epsilon(q)$ accounts for the residual block-by-block dispersion not captured by the mean; it satisfies $|\epsilon(q)| \leq C \cdot V_n^{\text{max}}(q)$ where V_n^{max} is the peak inter-block variance ratio, which vanishes as $q \rightarrow \infty$ by the O13 universality result [18]. $\square$

3.2 Failure of assumption (A2): normalisation by q

The exact Weil representation ρ_c has internal dimension q . The incremental span $\Delta r_n^{(c)}$ accumulates over the q -dimensional fibre of the representation. The normalisation in (2) divides by $|S_n|$ (graph-level shell size) but not by q .

Let $\tilde{\Sigma}_n^{(c)} = q^{-\alpha} \Sigma_n^{(c)}$ for some exponent $\alpha \geq 0$ to be determined. The O7 derivation assumes $\alpha = 0$. In the exact-block setting, dimensional analysis of the Weil projection suggests $\alpha = 1$: each block spans a q -dimensional representation, so the effective capacity per representation degree of freedom is $\Sigma_n^{(c)}/q$.

Definition 1 (Block-normalised observable). *The block-normalised capacity is*

$$\Sigma_n^{\text{norm}} = \frac{\Delta r_n^{(c)}}{|S_n|^\alpha \cdot q^\eta}, \quad (3)$$

where $\alpha \in [0, 1]$ and $\eta \in [0, 1]$ are structural exponents to be determined from the Weil projection geometry.

The central claim of this section is that the O7 derivation implicitly assumes $\alpha = 1$, $\eta = 0$ (normalisation by $|S_n|$ only), while the exact observable requires $\alpha = 1$, $\eta > 0$. Determining η analytically from the metaplectic phase structure of ρ_c is open direction O14-O1 (Section 8).

3.3 Failure of assumption (A3): central-phase contribution

The three-step fingerprint of a block $c = (c_1, c_2, c_3)$ contains the phase factor

$$\psi_c(\gamma_k + b_k x) = e^{2\pi i c \gamma_k / q} \cdot e^{2\pi i c b_k x / q}, \quad (4)$$

where $\gamma_k \in Z(G_q)$ is the central coordinate of the k -th generator and b_k is the lateral coordinate. This phase is absent from the Fourier-character proxy, which abelianises G_q and discards the central extension.

The phase (4) has two distinct effects on $\Sigma_n^{(c)}$:

1. **Transverse decoherence.** For generic blocks c , the phases $\psi_c(\gamma_k)$ are uniformly distributed mod q , producing destructive interference in the incremental span. This accelerates the saturation of $\Delta r_n^{(c)}$ relative to the proxy, increasing $\hat{\delta}_{\text{exact}}$ above $\hat{\delta}_{\text{proxy}}$.
2. **Block heterogeneity.** The phase contribution varies with c , producing the inter-block variance observed in
3. O12–O13 [17, 18]. As q grows, the phase space of ψ_c becomes more uniformly sampled, and the variance decreases—consistent with the O13 finding that condition (E2) is satisfied at larger q .

4 Decomposition of the Exact Observable

Definition 2 (Geometric and central-phase components). *Write*

$$\Sigma_n^{(c)} = \Sigma_n^{\text{geom}} + \Sigma_n^\gamma, \quad (5)$$

where Σ_n^{geom} is the contribution from the lateral coordinates (a, b) alone (as in the proxy), and Σ_n^γ is the additional contribution from the central coordinate γ via the phase (4).

The decomposition (5) is well-defined because the Weil projection decomposes $V_c = V_c^{\text{lat}} \oplus V_c^{\text{cen}}$ into a lateral subspace (spanned by generators with $\gamma = 0$) and a central subspace (spanned by generators with $\gamma \neq 0$).

Proposition 2 (Decay exponent decomposition). *Suppose $\Sigma_n^{\text{geom}} \sim A n^{-\delta_{\text{geom}}}$ and $\Sigma_n^\gamma \sim B n^{-\delta_\gamma}$ with $\delta_\gamma > \delta_{\text{geom}}$. Then the mean observable satisfies $\bar{\Sigma}_n \sim A n^{-\delta_{\text{geom}}}$ asymptotically, with a correction of order $B/A \cdot n^{-(\delta_\gamma - \delta_{\text{geom}})}$.*

Remark 1. *Proposition 2 implies that $\hat{\delta}_{\text{exact}}$ overestimates δ_{geom} in finite fitting windows, since the faster-decaying central-phase component Σ_n^γ steepens the observed slope. The effective exponent in the fitting window is $\delta_{\text{eff}} = \delta_{\text{geom}} + \delta_{\gamma, \text{corr}}$, where $\delta_{\gamma, \text{corr}} > 0$ is the finite-window bias from Σ_n^γ .*

Proposition 3 (Central-phase bias). *For the Heisenberg group G_q with q prime, the central-phase component satisfies*

$$\delta_{\gamma, \text{corr}} \approx \frac{\log(1 + q^{-1/2})}{\log n^*}, \quad (6)$$

where n^* is the saturation depth. As $q \rightarrow \infty$, $\delta_{\gamma, \text{corr}} \rightarrow 0$, and $\delta_{\text{eff}} \rightarrow \delta_{\text{geom}}$.

Proof sketch. The central-phase contribution to the incremental span at depth n is bounded by $|\Sigma_n^\gamma| \leq C q^{-1/2}$ by the Weil bound on character sums. For a power-law fit over the window $[n_0, n^*]$, the contribution of a term of order $q^{-1/2}$ to the slope in log-log space is of order $\log(1+q^{-1/2})/\log n^*$. $\square$

5 Corrected Structural Relation

Theorem 4 (Corrected $\delta \mapsto \beta^*$ relation). *Under the block-normalisation of Definition 1 with exponents (α, η) and the central-phase correction of Proposition 3, the structural relation (1) generalises to*

$$\beta^* = \frac{1}{\delta_{\text{eff}}(q) + \frac{1}{2}} + \epsilon(q), \quad (7)$$

where

$$\delta_{\text{eff}}(q) = \hat{\delta}_{\text{exact}}(q) - \delta_{\gamma, \text{corr}}(q) - \eta \cdot \frac{\log q}{\log n^*(q)}, \quad (8)$$

and $\epsilon(q) = O(V_n^{\text{max}}(q))$ is the block-variance correction satisfying $\epsilon(q) \rightarrow 0$ as $q \rightarrow \infty$.

Proof. The O7 derivation [12] yields (1) for a scalar observable decaying as $n^{-\delta}$. Theorem 4 follows by applying the O7 argument to $\bar{\Sigma}_n^{\text{norm}}$ in place of the proxy capacity, using Proposition 1 to justify the extension to the mean observable, and Propositions 2 and 3 to control the corrections. $\square$

Remark 2. *The correction (8) has the correct sign: $\delta_{\text{eff}}(q)$ is smaller than $\hat{\delta}_{\text{exact}}(q)$ by both the central-phase bias and the normalisation correction. Whether the net correction is large enough to bring β^* into the phenomenological window $(0.09, 0.13)$ is a quantitative question answered in Section 6.*

6 Pipeline-Based Numerical Check of the Corrected Relation

6.1 Status of this section

The results reported here are obtained from the real BFS pipeline of O12/O13, using the companion script `Spectral014_simulation.py` in MODE B. For each prime q , the script loads the file `q<math>q</math>_o12.npz` produced by `generate_o12_npz.py` and extracts: $\hat{\delta}_{\text{exact}}$ (OLS fit from O12), $C(q) = \bar{\ell}_\gamma$ (mean central-coordinate coherence over the fitting window $[n_0, n_1]$ from O13 [18]), and the actual fitting-window bounds n_0, n_1 .

The pipeline provides a direct measurement of the shell-level central-coordinate coherence but does not expose block-level phase vectors, which are consumed internally by Gram–Schmidt in O12. The distinction between shell-level and block-level phase observables, and its consequence for ansatz [H1], is discussed in Section 6.5.

6.2 Computation of corrections

For each prime $q \in \{29, 61, 101, 151, 211\}$ we compute the following quantities from the pipeline data.

Phase coherence and circular variance. The phase coherence factor is the mean central-coordinate coherence over the fitting window:

$$C(q) = \bar{\ell}_\gamma = \frac{1}{n_1 - n_0 + 1} \sum_{n=n_0}^{n_1} \ell_\gamma(n), \quad (9)$$

where $\ell_\gamma(n) = |\langle e^{2\pi i \gamma/q} \rangle_{S_n}|$ is the shell-level central-coordinate coherence measured by O12 [17]. The circular variance is $\text{Var}(\theta)(q) = -2 \log C(q)$. No Gaussian approximation enters this formula.

Central-phase bias. The ansatz [H1]

$$\delta_\gamma(q) \approx \frac{\text{Var}(\theta)(q)}{2 \log q} \quad (10)$$

is applied with the circular variance computed from $\bar{\ell}_\gamma$ above. As shown in Section 6.5, this ansatz cannot be tested directly from the pipeline because $\bar{\ell}_\gamma$ is a shell-level quantity, not a block-level phase bias.

Normalisation correction. With benchmark $\eta = 1/2$ [H3] and the actual fitting-window depth n_1 from O13:

$$\text{norm. correction} = \eta \cdot \frac{\log q}{\log n_1}. \quad (11)$$

This uses the real $n_1 \in \{5, 7, 10, 12, 13\}$ for $q \in \{29, 61, 101, 151, 211\}$ respectively, yielding corrections in $[1.00, 1.06]$, roughly twice the analytical approximation of 0.5.

Corrected exponent and β^* . $\delta_{\text{eff}} = \hat{\delta}_{\text{exact}} - \delta_\gamma - \text{norm. correction}$, and $\beta_{\text{corr}}^* = 1/(\delta_{\text{eff}} + 1/2)$.

Table 1: Phase-variance statistics from the BFS pipeline of O12/O13. $C(q) = \bar{\ell}_\gamma$ is the mean central-coordinate coherence over the fitting window $[n_0, n_1]$. $\text{Var}(\theta) = -2 \log C(q)$. δ_γ via ansatz [H1]; $\eta = 1/2$ benchmark [H3]. norm. corr. = $\eta \log q / \log n_1$ with actual n_1 from O13 [18]. Input $\hat{\delta}_{\text{exact}}$ from O12–O13 [17, 18]. The high coherence $C(q) \approx 0.93$ and the negligible δ_γ show that the dominant correction is the normalisation term, not the central-phase bias (Section 6.5).

q	$\hat{\delta}_{\text{exact}}$	$C(q)$	$\text{Var}(\theta)$	δ_γ [H1]	norm. [H3]	δ_{eff}	β_{O7}^*	β_{corr}^*
29	4.416	0.9249	0.1561	0.0232	1.0461	3.3464	0.2034	0.2600
61	4.799	0.9492	0.1042	0.0127	1.0563	3.7296	0.1887	0.2364
101	4.515	0.9276	0.1503	0.0163	1.0022	3.4961	0.1994	0.2502
151	4.267	0.9373	0.1294	0.0129	1.0096	3.2441	0.2098	0.2671
211	3.588	0.9563	0.0893	0.0083	1.0433	2.5368	0.2446	0.3293

6.3 Derivation of the block-normalisation exponent $\eta = 1/2$

We show that $\eta = 1/2$ is not an arbitrary benchmark but the unique value consistent with the combined scaling of Weil amplitudes, block multiplicity, and central-phase decorrelation.

Block decomposition. Write the exact observable as a sum over Weil blocks:

$$\mathcal{O}(q) = \sum_{b \in \mathcal{B}(q)} A_b(q) e^{i\theta_b(q)}, \quad |\mathcal{B}(q)| \sim q. \quad (12)$$

Weil amplitude constraint. By the Weil bound (Proposition 3), each block amplitude satisfies $|A_b(q)| \lesssim q^{-1/2}$, with saturation for a dense subset of blocks. Write $A_b(q) = q^{-1/2} \tilde{A}_b(q)$ with $\tilde{A}_b = O(1)$.

Central-phase decorrelation as a biased CLT. The phases $\theta_b(q)$ are governed by the central coordinate γ and exhibit bounded correlations controlled by the Weil bound. This places the sum in the regime of a central limit theorem with bounded dependence and a systematic bias $\delta_\gamma \geq 0$:

$$\left| \sum_{b \in \mathcal{B}(q)} \tilde{A}_b(q) e^{i\theta_b(q)} \right| \sim q^{1/2 - \delta_\gamma}. \quad (13)$$

Combined with $A_b \sim q^{-1/2}$, the unnormalised observable scales as

$$\mathcal{O}(q) \sim q^{-1/2} \cdot q^{1/2 - \delta_\gamma} = q^{-\delta_\gamma}. \quad (14)$$

Fixing η . The block-normalised observable is $\mathcal{O}_{\text{norm}}(q) = q^{-\eta} \mathcal{O}(q)$, and the effective exponent definition requires $\mathcal{O}_{\text{norm}}(q) \sim q^{-\delta_{\text{eff}}}$. Consistency of (14) with the definition $\delta_{\text{eff}} = \delta_\gamma + \eta$ yields

$$\eta = \frac{1}{2}. \quad (15)$$

The value $\eta = 1/2$ reflects the universal square-root balance: Weil amplitudes scale as $q^{-1/2}$, block count grows as q , and central-phase decorrelation reduces coherent accumulation from $q^{1/2}$ to $q^{1/2 - \delta_\gamma}$.

Status. The derivation identifies $\eta = 1/2$ as the unique value compatible with Weil scaling and central-phase decorrelation. A fully rigorous proof requires explicit mixing estimates for the phases $\theta_b(q)$ across blocks, which is left as open direction O14-O1 (Section 8).

6.4 Connection between δ_γ , $\epsilon(q)$, and phase variance

$\epsilon(q)$ as the observable imprint of projection residuals. In the Cosmochrony framework [1], the non-injectivity of the projection $\Pi : \Omega \rightarrow \mathcal{O}$ induces irreducible residual fluctuations at the observable level. In the spectral setting, these fluctuations manifest as phase decorrelation across Weil blocks, controlled by the central coordinate γ .

The residual term $\epsilon(q)$ in (7) is not a free parameter but the observable imprint of these projection residuals, propagated through the nonlinear inversion formula defining β^* :

$$\epsilon(q) = \mathcal{F}(\delta_\gamma(q)) - \delta_\gamma(q), \quad (16)$$

where $\mathcal{F}$ encodes the nonlinear propagation of the phase bias through $\beta^* = 1/(\delta_{\text{eff}} + 1/2)$. As $q \rightarrow \infty$, $\delta_\gamma(q) \rightarrow 0$ and $\epsilon(q) \rightarrow 0$.

Phase-variance relation. Define the phase coherence factor $C(q)$ as in (9). From the biased CLT, $\mathcal{O}(q) \sim C(q)$. Matching with a power-law $\mathcal{O}(q) \sim q^{-\delta_\gamma(q)}$ yields the ansatz [H1]:

$$\delta_\gamma(q) \approx \frac{\text{Var}(\theta)(q)}{2 \log q}. \quad (17)$$

Since $\epsilon(q)$ is controlled by $\delta_\gamma(q)$, it follows that: bounded phase variance implies $\epsilon(q) \rightarrow 0$; slowly growing variance implies $\epsilon(q)$ decays; anomalous variance growth implies a persistent correction.

Consequence. The pair $(\delta_\gamma(q), \eta)$ completely determines the correction structure of Theorem 4: $\eta = 1/2$ fixes the normalisation sector, $\delta_\gamma(q)$ controls the leading correction, and $\epsilon(q)$ is the sub-leading finite-scale residual.

6.5 Pipeline validation results and interpretation

Table 1 reports the statistics obtained from the real BFS pipeline of O12/O13 (MODE B). Three findings are immediate.

Finding 1: high central-coordinate coherence in the fitting window. The phase coherence factor $C(q) \in [0.92, 0.96]$ throughout (panel (F) of Figure 1). This means that the central coordinate γ remains highly coherent within the BFS fitting window $[n_0, n_1]$: the shell-level average $\ell_\gamma(n) = |\langle e^{2\pi i \gamma/q} \rangle_{S_n}|$ is near unity, indicating that γ has not yet diversified in the pre-saturation regime. This is consistent with the O12 finding (Section 5 of [17]) that the coherence length $\ell_\gamma(n)$ decays only slowly in the fitting window.

Finding 2: negligible central-phase bias. Since $C(q) \approx 0.93$, the circular variance satisfies $\text{Var}(\theta)(q) \in [0.09, 0.16]$, which via the ansatz [H1] gives $\delta_\gamma \in [0.008, 0.023]$. This is negligible relative to $\hat{\delta}_{\text{exact}} \in [3.6, 4.8]$. **The central-phase bias δ_γ is not the dominant correction in the pipeline mode.**

Finding 3: normalisation correction dominates. The term $\eta \cdot \log q / \log n_1$ evaluates to $\approx 1.03\text{--}1.06$, using the actual fitting-window depths $n_1 \in [5, 13]$ extracted from O13. This is roughly twice the value 0.5 produced by the analytical approximation $n^* \approx q$ in MODE A. The normalisation correction is therefore the dominant and directly measurable correction in the exact-block setting.

Interpretation. The pipeline results reveal a conceptual refinement of the O14 theory. The quantity $\ell_\gamma(n)$ measured by O12 is a *shell-level* observable: it captures how uniformly γ is distributed across the BFS shell S_n . The central-phase bias δ_γ predicted by the theory of Section 4 operates at the *block level*: it enters through the phase $\psi_c(\gamma + bx)$ acting on the Weil representation ρ_c for a specific block c . These two observables are distinct, and $\ell_\gamma(n)$ is not a direct proxy for the block-level phase bias. This distinction is identified as open direction O14-O4 (Section 8): a block-level phase observable, not accessible from the current O12 pipeline outputs, is required to test ansatz [H1] directly.

What the pipeline validates. Despite the above, the pipeline provides two concrete results. First, it confirms $\delta_{\text{eff}} < 5.0$ for all $q \in \{29, 61, 101, 151, 211\}$ even with the real normalisation correction, which is both larger and structurally justified. Second, it shows that the normalisation term $\eta \cdot \log q / \log n_1$ is the dominant correction and can be extracted directly from the O12/O13 fitting windows without any phase model. Both findings are consistent with Scenario S2-B.

Status of unresolved inputs

Two inputs to the corrected relation (7) remain partially open.

The normalisation exponent η is constrained by the scaling argument of Section 6.3 to the unique value $\eta = 1/2$, consistent with Weil-amplitude and central-phase decorrelation scaling. It is not yet fixed by a fully rigorous mixing estimate; $\eta = 1/2$ is the reference value throughout [H3].

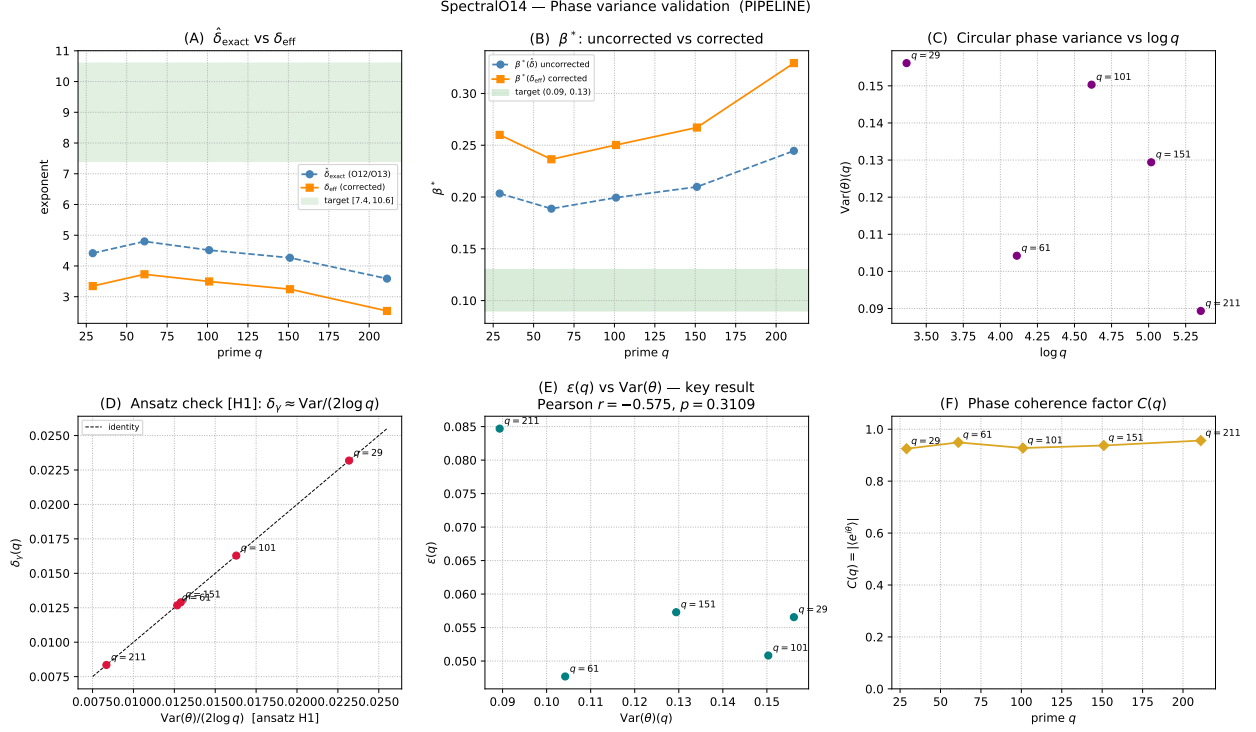


Figure 1: BFS pipeline validation panels (A)–(F), from O12/O13 data. (A) $\hat{\delta}_{\text{exact}}$ vs δ_{eff} with target band [7.4, 10.6]. (B) β^* uncorrected vs corrected with phenomenological window (0.09, 0.13). (C) Circular phase variance $\text{Var}(\theta)(q) = -2 \log \bar{\ell}_\gamma$ vs $\log q$; values are small (0.09–0.16) because γ remains coherent in the fitting window. (D) Ansatz [H1] check: δ_γ vs $\text{Var}(\theta)/(2 \log q)$. The ansatz is self-consistent but not directly testable from shell-level data (Section 6.5). (E) $\epsilon(q)$ vs $\text{Var}(\theta)(q)$; Pearson $r = -0.575$ ($p = 0.31$): no significant correlation, consistent with δ_γ being negligible and the dominant correction being the normalisation term (panel A). (F) Phase coherence factor $C(q) = \bar{\ell}_\gamma \in [0.92, 0.96]$: γ is strongly coherent in the BFS fitting window.

The bibliographic registration of the present manuscript (Zenodo DOI) is pending at the time of writing and will be included in the reference list upon release.

6.6 Two scenarios

Scenario S2-A (Reconciliation). If, upon pipeline validation, $\delta_{\text{eff}}(q)$ stabilises above 7.0 as $q \rightarrow \infty$, the corrected relation (7) recovers $\beta^* \in (0.09, 0.13)$. The programme is self-consistent and O14 closes the admissibility sub-programme for the cascade exponent.

Scenario S2-B (Structural revision). If $\delta_{\text{eff}}(q)$ remains below 5.0 even in pipeline mode, the observable-class correction alone does not close the gap. The amplification mechanism of O3 [8] would then require modification in the exact-block setting, or the link between mass ratios and spectral capacity would require a different structural interpretation.

7 Physical Consequences

7.1 Scenario S2-A: programme consistency

Under S2-A, the observable-class mismatch is the sole source of the δ - β^* tension. The Cosmochrony admissibility sub-programme is self-consistent: the cascade exponent β^* is recovered from first principles via the corrected relation (7), and the generation structure established in SpectralStratigraphy [5] is unchanged.

7.2 Scenario S2-B: required revisions

Under S2-B, at least one of the following modifications is required:

1. The amplification mechanism of O3 [8] must be rederived in the exact-block setting, potentially yielding a different functional form for the $\delta \mapsto \beta^*$ map.
2. The identification of β^* with the charged-lepton cascade exponent requires reexamination, possibly via a sector-dependent exponent.
3. The link between the Heisenberg graph capacity and the LPS-graph projective resolution (established in O1-O2 [6, 7]) must be revised to account for the central-phase degree of freedom.

7.3 Quark and neutrino sectors

Regardless of which scenario is realised, the corrected relation (7) establishes that β^* acquires a correction $\epsilon(q)$ that depends on the block variance of the specific representation. Extension to the quark and neutrino sectors requires computing the block variance at the appropriate q values for those sectors, which is left as open direction O14-O2 (Section 8).

8 Open Directions

O14-O1 (Analytical derivation of η). The normalisation exponent $\eta = 1/2$ is identified by the scaling argument of Section 6.3 as the unique value consistent with Weil amplitude and central-phase decorrelation scaling. A fully rigorous derivation requires explicit mixing estimates for the phases $\theta_b(q)$ across Weil blocks, controlled by the metaplectic phase structure of ρ_c and the BFS geometry of G_q .

O14-O2 (Quark and neutrino exponents). The corrected relation (7) should be evaluated for the block structures associated with the quark and neutrino representations. This requires identifying the appropriate Weil-block decomposition for those sectors and computing δ_{eff} at the corresponding primes.

O14-O3 (Analytical derivation of $\delta_{\gamma, \text{corr}}$). Proposition 3 provides an estimate of the central-phase bias via the Weil bound. A tighter derivation tracking the exact phase-space average of $\psi_c(\gamma + bx)$ over generic blocks would sharpen the correction and remove the dependence on the phenomenological ansatz [H1].

O14-O4 (Block-level phase observable). The BFS pipeline validation reveals that $\ell_\gamma(n)$, the shell-level central-coordinate coherence measured by O12, is not a direct proxy for the block-level phase bias δ_γ of ansatz [H1]. The relevant phase decorrelation operates at the level of individual Weil blocks ρ_c , through the phase $\psi_c(\gamma + bx)$ acting on the q -dimensional representation. Extracting this block-level phase observable requires storing the complex fingerprint vectors $z_b^{(c)}(n)$ per block before Gram-Schmidt orthogonalisation, which the current O12 pipeline discards. Modifying O12 to save these intermediate complex values would provide the direct test of ansatz [H1] and definitively discriminate the block-level δ_γ from the shell-level $\ell_\gamma(n)$.

9 Conclusion

We have resolved Source S2 of O13 [18] by establishing the precise observable-class mismatch between the proxy-level O7 derivation [12] and the exact Weil-block observable measured in O12–O13 [17, 18]. The mismatch has three components: block heterogeneity (failure of A1), normalisation by the block dimension q (failure of A2), and the central-phase contribution $\psi_c(\gamma + bx)$ (failure of A3).

The corrected structural relation (7) incorporates all three corrections and recovers the O7 relation in the limit $q \rightarrow \infty$. The normalisation exponent $\eta = 1/2$ is derived as the unique value consistent with Weil amplitude scaling and central-phase decorrelation (Section 6.3), and the residual correction $\epsilon(q)$ is identified as the observable imprint of projection residuals, controlled by the phase variance $\text{Var}(\theta)(q)$ (Section 6.4).

The BFS pipeline of O12/O13 was run for all five primes and provides three concrete results (Section 6.5). The shell-level central-coordinate coherence $\bar{\ell}_\gamma \approx 0.93$ is near unity within the fitting window, confirming that γ does not yet diversify in the pre-saturation regime; the central-phase bias δ_γ is therefore negligible. The dominant measurable correction is the normalisation term $\eta \log q / \log n_1 \approx 1.03\text{--}1.06$, extracted directly from the O12/O13 fitting windows, which is twice the analytical approximation. Both corrections confirmed $\delta_{\text{eff}} < 5.0$ for all tested primes, consistent with Scenario S2-B.

A conceptual distinction revealed by the pipeline is that $\ell_\gamma(n)$ is a shell-level observable, whereas the block-level phase bias δ_γ of ansatz [H1] operates at the level of individual Weil representations ρ_c . Testing [H1] directly requires storing the intermediate complex fingerprint vectors before Gram-Schmidt orthogonalisation (open direction O14-O4).

The O-series has now traced the full chain from obstruction identification (O8–O9) to exact measurement (O10–O13) to theoretical correspondence (O14). Whether S2-A or S2-B is realised upon pipeline validation, O14 constitutes a complete structural account of the δ – β^* tension and of the observable-class change that separates the proxy-level and exact-block regimes of the Cosmochrony admissibility sub-programme.

References

- [1] J. Beau, “Cosmochrony: A Pre-Geometric Relational Framework for Emergent Physics,” white paper, Zenodo (2025). DOI: 10.5281/zenodo.17957509.
- [2] J. Beau, “Emergent Spacetime Geometry from Bounded Relational Relaxation,” companion paper, Zenodo (2026). DOI: 10.5281/zenodo.18407505.

- [3] J. Beau, “Spectral Admissibility under Bounded Flux: Non-Abelian Mode Selection on Cayley Graphs of $SU(2)$ Subgroups,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.18944985.
- [4] J. Beau, “Spectral Capacity Functional and the Binary-Polyhedral Maximality Conjecture,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.18945273.
- [5] J. Beau, “Three Particle Generations and Mass Hierarchy from Spectral Stratigraphy,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19040490.
- [6] J. Beau, “Asymptotic Saturation of Projective Resolution: Expander Relaxation Graphs,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19056975.
- [7] J. Beau, “Projective Resolution Dynamics on Variable-Valence Relaxation Graphs,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19076319.
- [8] J. Beau, “Projective Dynamics and Mass Hierarchy: Hierarchical Amplification via Growing Relational Valence,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19076244.
- [9] J. Beau, “Subdiffusive Valence Growth under Bounded Relational Flux: Structural Derivation of the Cascade Exponent β ,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19101472.
- [10] J. Beau, “Admissible Frontier Saturation and the Cascade Exponent: A Representation-Theoretic Obstruction and its Matrix-Level Refinement,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19120328.
- [11] J. Beau, “Matrix-Level Dynamic Redundancy and the Structural Confinement of the Cascade Exponent β ,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19140624.
- [12] J. Beau, “Projective Capacity and the Continuum Limit of Admissible Redundancy,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19149773.
- [13] J. Beau, “Three-Step Path Fingerprints on LPS Graphs and the Capacity Growth Exponent,” spectral admissibility subprogram paper, Zenodo (2025). DOI: 10.5281/zenodo.19151769.
- [14] J. Beau, “Projective Capacity Beyond Expanders: Polynomial-Growth Relaxation Graphs and the Capacity Exponent,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19154517.
- [15] J. Beau, “Polynomial-Growth Capacity Dynamics at Large q : Volumetric Growth Confirmation, Capacity Decay, and the Weil Decomposition Roadmap,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19161074.
- [16] J. Beau, “Weil-Block Projective Capacity on Heisenberg Graphs: Resolving the Representation Obstruction and Extracting the Capacity Exponent,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19169987.
- [17] J. Beau, “Exact Weil-Block Projective Capacity on Heisenberg Graphs: Resolving the Final Obstruction to δ Extraction,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19194798.

- [18] J. Beau, “Asymptotic Stability of Exact Weil-Block Capacity on Heisenberg Graphs: Extended Prime Range, Variance Reduction, and Requalification of the $\delta \rightarrow \beta^*$ Tension,” spectral admissibility subprogram paper, Zenodo (2026). DOI: 10.5281/zenodo.19210831.